Session 5 — Analysis

Exercise sheet

These questions follow the session in order and start very gently. Two examples come back throughout: the forgetting curve $R(t) = R_0 e^{-t/\tau}$*, the proportion of a word list still recalled after a delay* t*, and the* learning curve $P(t)$*, performance climbing toward a ceiling with practice. Three numbers are worth having in front of you:* $e^{-1} \approx 0.37$, $e^{-2} \approx 0.14$, $e^{-3} \approx 0.05$*. Show your working, and if a question defeats you write* "I don't know" *— we will do it together.*

Exercise 1 Growth and decay — read the sign of a first.

(a) Does e^{3t} grow or decay? And e^{-3t}? And $e^{-t/5}$?

(b) Write $e^{-t/5}$ in the form e^{at}: give a, then give the time constant τ.

(c) Write $e^{-t/2}$, $e^{-t/10}$ and e^{-2t} in order, from the fastest decay to the slowest.

Exercise 2 *Recall of a learned word list decays as* $R(t) = R_0 e^{-t/\tau}$.

Take $R_0 = 0.8$ and $\tau = 3$ days.

(a) Compute $R(0)$, $R(3)$, $R(6)$ and $R(9)$.

(b) Sketch the curve roughly, marking those four points.

(c) Someone says "after 3 days the memory is gone". Why is that wrong, and what does $\tau = 3$ actually mean?

Exercise 3 Two participants forget with $\tau = 2$ days and $\tau = 8$ days, both starting from $R_0 = 1$.

(a) After 8 days, roughly how much does each still recall?

(b) Which of the two has the better memory, and is it the larger or the smaller τ?

Exercise 4 Differentiate, saying each time which rule you are using.

(a) $f(t) = t^2 - 4t + 7$;

(b) $g(t) = e^{3t}$;

(c) $h(t) = 5\,e^{-2t}$;

(d) $k(t) = 1/t + \sqrt{t}$.

Exercise 5 Base or exponent?

(a) Differentiate t^2 and e^{2t}. Which rule applies to each, and why?

(b) Someone writes $\frac{d}{dt}e^{-t/2} = \left(-\frac{t}{2}\right) e^{(-t/2)-1}$. What have they done wrong? Give the correct answer.

Exercise 6

(a) Differentiate $R(t) = 0.8\, e^{-t/4}$ and check that $R' = -R/4$.

(b) Say in words what the equation $R' = -R/4$ asserts about forgetting.

(c) Differentiate $P(t) = 100 - 60\, e^{-t/5}$. Is P rising or falling? Watch the two minus signs.

Exercise 7 Using the sign of the derivative.

(a) For $f(t) = t^2 - 4t + 7$: where is $f' = 0$? Check the sign of f' on either side and say whether it is a peak or a trough.

(b) For $f(t) = 6 + 4t - t^2$: same questions.

(c) For $P(t) = 100 - 60\, e^{-t/5}$: show $P' > 0$ for every t, and say what value P approaches.

Exercise 8 Compute, using the table of antiderivatives, and check each answer by differentiating it.

(a) $\int_0^3 (2t + 1)\, dt$;

(b) $\int_0^2 (3t^2 - 4t + 1)\, dt$;

(c) $\int_0^\infty e^{-2t}\, dt$;

(d) an antiderivative of $e^{-t/5}$.

Exercise 9

(a) Show that $\int_0^\infty e^{-t/\tau}\, dt = \tau$.

(b) A rate of responding is $r(t) = 4$ responses per minute for 10 minutes. Write the total number of responses as an integral and compute it.

(c) In one sentence: what is the relationship between a derivative and an integral?

Exercise 10 Partial derivatives: differentiate one letter at a time, treating the other as a constant.

(a) $f(x, y) = x^2 + 3xy + y$: compute $\partial f/\partial x$ and $\partial f/\partial y$.

(b) $f(x, y) = x^2 + 4y^2$: same.

(c) $f(x, y) = 5x + 2y - 7$: same. What do you notice?

(d) $f(x, y) = x\,\mathrm{e}^{-y}$: same.

Exercise 11

(a) For $f(x, y) = x^2 + 3xy + y$, evaluate $\partial f/\partial x$ at $(1, 0)$ and at $(1, 2)$.

(b) Why are the two answers different?

Exercise 12 *A* contour line *of f is the set of points where f takes one fixed value. An iso-performance curve — all the pairs (difficulty, practice) giving the same accuracy — is exactly that.*

(a) Sketch the contour lines $f = 1$ and $f = 4$ of $f(x, y) = x^2 + y^2$. What shape are they?

(b) Sketch the contour line $f = 4$ of $f(x, y) = x^2 + 4y^2$.

(c) On a contour line, does f change? Where on a contour map is the surface steepest?

Exercise 13 Vector-valued functions: differentiate one component at a time.

(a) $\boldsymbol{x}(t) = (2t,\ 3 - t)$: compute $\dot{\boldsymbol{x}}$ and the speed $\|\dot{\boldsymbol{x}}\|$. What shape is the path?

(b) $\boldsymbol{x}(t) = (t,\ t^2)$: compute $\dot{\boldsymbol{x}}$, then the velocity at $t = 0$ and at $t = 1$.

(c) $\boldsymbol{x}(t) = (t,\ \mathrm{e}^{-t})$: compute $\dot{\boldsymbol{x}}$ and give the velocity at $t = 0$.

Exercise 14 *In a mouse-tracking experiment the cursor position $(x(t), y(t))$ is recorded as the participant moves toward one of two response buttons. A path that bends toward the* unchosen *button is evidence that the two responses competed.*

A trajectory is $\boldsymbol{x}(t) = (t,\ t - t^2)$ for $t \in [0, 1]$.

(a) Where is the participant at $t = 0$ and at $t = 1$?

(b) Compute $\dot{\boldsymbol{x}}(t)$ and give the velocity at $t = 0$.

(c) At which t is y largest? What does that moment represent?

Exercise 15 For each, say whether the rate depends on *time* (integrate) or on *the quantity itself* (a differential equation).

(a) $\dot{x} = 3t^2$;

(b) $\dot{x} = -2x$;

(c) $\dot{x} = -2x + 5$;

(d) $\dot{x} = \mathrm{e}^{-t}$.

Exercise 16

(a) Solve $\dot{x} = -2x$ with $x_0 = 5$. Where does x go as t grows?

(b) Solve $\dot{x} = 0.5x$ with $x_0 = 2$. Where does x go?

(c) Check both answers by differentiating them.

Exercise 17 The equation $\dot{x} = ax + b$, with solution $x(t) = x^\star + (x_0 - x^\star)\mathrm{e}^{at}$ and $x^\star = -b/a$.

(a) For $\dot{x} = -x + 3$: find $x^\star$ by setting the rate to zero, give τ, and write $x(t)$ starting from $x_0 = 0$. Is it stable?

(b) For $\dot{x} = -x + 3$ starting from $x_0 = 10$: write $x(t)$ and describe it.

(c) Check by substitution that $x(t) = 3 + C\,\mathrm{e}^{-t}$ solves $\dot{x} = -(x - 3)$ for *every* constant C. Why do we need an initial condition?

Exercise 18 Someone proposes $-x^2/2 + 3x$ as the solution of $\dot{x} = -(x - 3)$.

(a) What operation did they perform?

(b) Give the one-line reason it cannot be a solution. *(Hint: what is the answer supposed to be a function of?)*

Exercise 19 *Performance climbs toward a ceiling, and the closer you get the slower you improve:* $\dot{P} = k\,(P_{\max} - P)$.

(a) Take $P_{\max} = 90$, $k = 0.2$ per session, $P_0 = 30$. Give $P^\star$ and τ.

(b) Write $P(t)$ and compute $P(5)$ and $P(15)$.

(c) Roughly how many sessions are needed to close 95% of the gap?

(d) In one sentence: what do the forgetting curve and the learning curve have in common?

Exercise 20 Stability without solving.

(a) For $\dot{x} = -x + 4$: find $x^\star$ and, without solving, say what happens starting from $x = 10$ and from $x = 0$.

(b) For $\dot{x} = 2x - 6$: find $x^\star$ and do the same. Is it stable?

(c) Fill in: the fixed point is stable when a is ___________ and unstable when a is ___________.